Ernesto Contreras Alarcon

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PROFESSIONAL SUMMARY

Software Engineering graduate (First Class Honours) with a backend and generative AI profile: Python, FastAPI, Node.js and SQL, LLM integration (OpenAI, Amazon Bedrock, Gemini), RAG systems and automated testing, on Azure (Azure AI Foundry) and AWS. At CEU I worked with teaching staff and real users on an AI academic advisor, using their feedback to improve its answers. Looking to join a software engineering team, preferably in AI, where I can keep learning and contribute from day one.

PROFESSIONAL EXPERIENCE

CEU – IT Innovation Department Feb. 2025 – Jun. 2025
AI Software Engineer Intern Valencia, Spain

- Built an AI Personal Academic Advisor integrating OpenAI APIs with institutional Snowflake data to give context-aware, verifiable answers to academic queries, piloted with real users.
- Worked with teaching staff and real users, using their feedback to iterate on XML-structured prompts: better response relevance, fewer hallucinations and lower token consumption.
- Contributed to a Python and SQL backend on Azure connecting institutional data with AI model inference.
- Optimised data retrieval through targeted SQL joins and query logic, reducing reported access times from days to sub-second responses.

FEATURED AI PROJECTS

Enterprise Hybrid RAG Engine API | *Python, FastAPI, PostgreSQL, pgvector, GitHub Actions, pytest* 2026

- Built a hybrid retrieval API combining semantic vector search and lexical search using PostgreSQL, pgvector, HNSW, and GIN indexes.
- Implemented Reciprocal Rank Fusion (RRF) for document ranking and an ingestion pipeline with batch processing, exponential-backoff retries, and source-citation parsing.
- Created pytest suites and GitHub Actions CI checks covering API endpoints, retrieval ranking, and vector database operations.

Ring TelemetryOps – AI-assisted Diagnostic API | *Python, FastAPI, OpenCV, Amazon Bedrock, pytest* 2026

- Built a FastAPI diagnostic agent for Ring video doorbells for the Amazon Developer Hackathon 2026 (Ring track): it analyses video quality with OpenCV (frame rate, bitrate, sharpness, banding) and correlates it with simulated device and network telemetry to identify likely faults.
- Designed reproducible fault scenarios and a deterministic retrieval pipeline that ranks troubleshooting procedures from a 15-article knowledge base, with a local fallback when the model is unavailable.
- Integrated Amazon Bedrock (Claude) for reasoning; delivered REST endpoints, a CLI, an interactive web dashboard, and pytest suites that run without credentials.

WakeMap – AI Location & Travel Platform | *Flutter, Node.js, Express, Gemini API* 2025–2026

- Developed a cross-platform Flutter prototype with location-aware alerts and OpenStreetMap integration for journey planning.
- Built a Node.js/Express backend proxy that keeps the Gemini API key server-side and generates travel guidance from user context, time, budget, preferences, and walking-distance constraints.

EDUCATION

York St John University 2025 – 2026
BSc (Hons) Software Engineering – First Class Honours (74.50%) York, United Kingdom

Universidad CEU Cardenal Herrera 2023 – 2025
Higher Technician in Multiplatform Application Development (DAM) – Grade: 8.5/10 Valencia, Spain

Hugging Face 2026
AI Agents Course (LangGraph, smolagents, LlamaIndex, MCP, Langfuse) Online

TECHNICAL SKILLS

Programming: Python, Java, TypeScript/JavaScript, Go (basic); object-oriented programming.
AI-assisted Development: Daily use of Claude Code, Codex and Antigravity across development and testing.
AI & LLMs: LLM APIs (OpenAI, Amazon Bedrock, Gemini), hybrid RAG, embeddings, vector search (PostgreSQL, pgvector), prompt engineering (XML-structured prompts), evaluation of response quality and token usage; LangGraph, smolagents, LlamaIndex, MCP, Langfuse (Hugging Face AI Agents course).
Cloud & DevOps: Azure (Azure AI Foundry), AWS (Amazon Bedrock), Docker/Docker Compose, Git (command line), GitHub Actions (CI), pytest.
Data & Backend: FastAPI, Node.js/Express, REST APIs, SQL, PostgreSQL, Snowflake, batch processing, retry/backoff.
Languages: Spanish (Native), English (C1, UK university graduate).